INSTITUTE OF AERONAUTICAL ENGINEERING
(Autonomous)
Dundigal, Hyderabad - 500 043

COMPUTER SCIENCE AND ENGINEERING

TUTORIAL QUESTION BANK

Course Title	**COMPUTER NETWORKS**				
Course Code	AITD04				
Program	B.Tech				
Semester	IV	CSE			
Course Type	Core				
Regulation	BT23				
Course Structure	Theory			Practical	
	Lecture	Tutorials	Credits	Laboratory	Credits
	3	-	3	-	-
Course Coordinator	Ms. B.K. Aishwarya, Assistant Professor				

COURSE OBJECTIVES:

The students will try to learn:

I	The fundamentals of computer networks, including network types, transmission media, and core models like TCP/IP and OSI.
II	Link-layer protocols, addressing, and error correction techniques for efficient communication.
III	Routing, addressing, congestion control, and QoS in the network layer.
IV	Transport protocols (UDP, TCP) and evaluate network performance.
V	Advanced concepts like DOCSIS, MPLS, and data center networking through practical lab exercises.

COURSE OUTCOMES:

After successful completion of the course, students should be able to:

CO 1	**Outline** the basic concepts of data communications including the key aspects of networking and their interrelationship, packet, circuit and cell switching as internal and external operations, physical structures, types, models, and internetworking	Understand
CO 2	**Make use of** different types of bit errors and the concept of bit redundancy for error detection and error correction	Understand
CO 3	**Identify** the suitable design parameters and algorithms for assuring quality of service and internetworking in various internet protocols.	Apply
CO 4	**Interpret** transport protocols (TCP, UDP) for measuring the network performance.	Analyse

CO 5	**Illustrate** the various protocols (FTP, SMTP, TELNET, EMAIL, and WWW) and standards (DNS) in data communications among networks.	Apply
CO 6	**Compare** various networking models (OSI, TCP/IP) in terms of design parameters and communication modes	Analyse

QUESTION BANK:

Q.No	QUESTION	Taxonomy	How does this subsume the level	CO's
MODULE I				
INTRODUCTION				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1.	Imagine you are the IT manager for a mid-sized company. Your current network setup is a mix of old and new technologies, and you are tasked with redesigning it to improve efficiency and scalability. Describe the steps you would take to evaluate your current network and plan for the upgrade. Which network type (LAN, MAN, WAN) would you choose for different departments, and why?	Understand	This would require the learner to understand the concept of network. Then recall the types of networks.	CO6
2.	With a network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?	Apply	This would require the learner to recall network concepts and illustrate network bandwidth and then identify throughput of this network.	CO1
3.	Differentiate OSI reference model with the TCP/IP reference model.	Understand	This would require the learner to: recall the concept of data communication model	CO1
4.	Explain the significance of Switching. What are different switching techniques used in computer networks? Discuss.	Understand	This would require the learner to: recall the concept of switching	CO1
5.	Analyze the impact of switching methods (circuit-switched vs. packet-switched) on the efficiency and quality of	Apply	This would require the learner to: recall the concept of transmission media	CO6

	service for a video conferencing application. What are the pros and cons of each method in terms of latency, bandwidth utilization, and reliability?			
6.	Discuss briefly about the original ARPANET design	Understand	This would require the learner to: recall the concept of history of internet .	CO1
7.	Suppose you are part of a team tasked with developing a new internet standard for a more secure communication protocol. Outline the steps your team would follow from conception to implementation. Identify the key organisations you would collaborate with and the challenges you might face during this process.	Apply	This would require the learner to recall the internet standard concept and demonstrate transmission medium and apply the procedure to Find out channel capacity	CO6
8.	Assume the difference between circuit switching and packet switching. Assume the link's rate is 2 Mbps and users are generating data at a rate of 100 Kbps when busy. Users are busy only a. What is the maximum number of users that a circuit switching architecture can support simultaneously?	Understand	This would require the learner to understand switching concept Then recall the formula and its parameters. Then assigning correct values for the parameters and solving.	CO1
9.	Trace the development of the internet from ARPANET to the modern day, highlighting at least three critical technological advancements or milestones. Discuss how each milestone contributed to the growth and functionality of the internet as we know it today.	Understand	This would require the learner to recall the signal concept and demonstrate the transmission medium and apply the procedure to find out Attenuation	CO1
10.	A university campus network consists of multiple buildings spread across several acres. The network needs to support both academic and administrative functions, including high-speed internet access, secure data transfer, and extensive	Understand	This would require the learner to understand the concept of network. Then recall the types of networks.	CO6

	Wi-Fi coverage. Propose a comprehensive network design, detailing the types of networks (LAN, MAN, WAN) and topologies you would implement. Justify your choices.			
PART-B LONG ANSWER QUESTIONS				
1.	Define switching and Explain packet switching.	Understand	—-	CO1
2.	Illustrate the differences between the OSI and TCP/IP Reference Models.	Understand	This would require the learner to recall the concept of OSI and TCP/IP model then explain the Differences of OSI and TCP/IP Reference Models.	CO1
3.	Define computer networks? Describe various types of network topologies in computer networks. Also, discuss various advantages and disadvantages of each topology.	Understand	—-	CO1
4.	Define switching. Explain circuit switching.	Understand	This would require the learner to recall switching concepts. Different switching Techniques then explain circuit switching	CO1
5.	Illustrate the differences between baseband transmission and broadband transmission.	Understand	This would require the learner to recall the concept of baseband transmission and broadband transmission and then compare between baseband transmission and Broadband transmission.	CO1
6.	Summarize TCP/IPModel. Explain the functions protocols and services of each layer. Compare it with the OSI Model.	Understand	This would require the learner to recall concept of TCP/IP model, explain the functions and protocols and services of each layer then compare to OSI Model	CO1
7.	With a neat sketch and Explain ISO/OSI reference model.	Understand	This would require the learner to recall concept of ISO/OSI reference model then explain the Layers of the ISO/OSI reference model.	CO1
8.	Discuss and Compare various types of networks.	Understand	This would require the learner to recall concept of computer networks, Then explain the various types of networks.	CO1
9.	List out and Explain the applications of Computer Networks.	Understand	—-	CO1
10.	Define OSI Model. Explain the functions and protocols and	Understand	—-	CO1

	servicesofeachlayer.			
11.	Explainthefollowing:-a) LAN b) MAN c) WAN d) ARPANET	Understand	This would require the learnerto recall conceptof computer networks then explain the various typesof networks (LAN , MAN,WAN and ARPANET)	CO1
12.	DiscusshowOSIandISO relatedtoeachother.	Understand	This would require the learnertorecallconceptof ISO and OSI reference model then explain the Relationof ISO and OSI reference model.	CO1
13.	Illustrate some of the factors that determine whether a unification system is a LAN or WAN.	Understand	This would require the learner to recall the concept of factors unification system in a LAN or WAN .then explain the factors that determine whether a unificationsystem isa LAN or WAN.	CO1
14.	Definebitrateandexplain the factorsthateffectsthe bit rate	Understand	—-	CO1
15.	Which characteristics affect the qualityofserviceoffered bya network? Justifyyour answer with properexample	Understand	This would require the learner to recall conceptsof datarateof achannelthen explain Shannon Capacity with example	CO1
16.	Discuss briefly about the original ARPANET design	Understand	This would require the learner to recall concepts of internet of history	CO1
17.	Explain the significance of Switching? What are different switching techniquesusedin computer networks? Discuss.	Understand	This would require the learner to recall concepts of switching	CO1
PART-C SHORT ANSWER QUESTIONS				
1.	List two disadvantages of twisted pair cables.	Remember	—-	CO1
2.	Define packet switching.	Remember	—-	CO1
3.	Listoutthetwoadvantages and thetwodisadvantages ofbus topologyinnetwork.	Remember	—-	CO1
4.	Listoutthetwoadvantages of layering principles in computer networks.	Remember	—-	CO1
5.	Define between baseband transmission and broadband transmission.	Remember	—-	CO1
6.	Define network.	Remember	—-	CO1
7.	Listdifferenttypesof networks.	Remember	—-	CO1
8.	Illustrate why protocols are needed.	Remember	This would require the learner to recall protocol concepts and explain the need of protocols.	CO1
9.	List the factors to measure the performance of network.	Remember	—-	CO1

10.	What is meant by topology? Name some popular topologies.	Remember	—-	CO1
11.	Define switching.	Remember	—-	CO1
12.	Define Why standards are needed.	Remember	—-	CO1
13.	Explain the importance about MAN.	Remember	This would require the learner to recall concepts of Networks then explain the importance of MAN.	CO1
14.	Describe the Noise.	Remember	This would require the learner to recall and outline the concept of Noise	CO1
15.	Explain a short note on WAN.	Remember	This would require the learner to recall the concepts of Networks then outline about WAN.	CO1
16.	Define Distortion.	Remember	—-	CO1
17.	Explain briefly internet history.	Remember	This would require the learner to recall the concept of internet history then explain history of internet.	CO1
18.	List the types of the Transmission mediums.	Remember	—-	CO1
19.	Illustrate the importance of LAN.	Remember	This would require the learner to recall concepts of Networks then explain he importance of LAN.	CO1
20.	Define Attenuation.	Remember	—-	CO1
21.	Define Shannon Capacity.	Remember	—-	CO1
22.	List out the four basic topologies.	Remember	—-	CO1
23.	List two advantages and disadvantages of computer networks.	Remember	—-	CO1
24.	List out the Layers of the OSI model.	Remember	—-	CO1
25.	List out the layers of the TCP/IP reference model.	Remember	—-	CO1
MODULE II				
DATALINK LAYER				
PART-A PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1.	Imagine the network design you proposed for VLAN implementation results in unexpected network slowdowns and intermittent connectivity issues. Identify potential problems that could arise from VLAN misconfigurations or inadequate design choices. How would you troubleshoot and	Apply	This would require the learner to recall the concept of the data link layer and identify the issues.	CO6

	resolve these issues?			
2.	Illustrate a network with one primary and four secondary stations uses polling. The size of a data frame is 1000 bytes. The size of the poll, ACK and NAK frames are 32 bytes each. Each station has 5 frames to send. How many total bytes are exchanged if there is no limitation on the number of frames a station can send in response to a poll?	Understand	This would require the learner to recall the concepts of polling and frames and then explain the How many total bytes are exchanged if there is no limitation on the number of frames	CO2
3.	Find CRC for P = 110011 and M = 1100011	Apply	This would require the learner to recall the concept of CRC .Then understand how to transfer the error free data by applying the XOR and AND operation	CO2
4.	One hundred stations on a pure ALOHA network share a 1- Mbps channel. If frames are 1000 bits long, find the throughput if each station is sending 10 frames/sec	Apply	Learner to recall the pure ALOHA and then apply &find the throughput of each station	CO2
5.	Calculate the hamming distance for each of the following code words i. d(10000,01000) ii. d(10101, 10010) iii. d(1111,1111) iv. d(0000,0000)	Apply	Learner to recall the hamming code and then apply &find hamming distance of given code word.	CO2
6.	Describe the purpose and function of cyclic redundancy checks (CRC) inerror detection. Howdoes CRC ensure data integrity during transmission?	Understand	Learner to recall the concept of CRC and its functions.	CO2
7.	Assuming even parity, find the parity bit for each of the following data units.a. 1001011 b. 0001100 c. 1000000 d. 1110111	Apply	This would require the learner to recall the concept of CRC .Then understand how to transfer the error free data by applying the XOR	CO2
8.	Given the dataword 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division).	Apply	This would require the learner to recall the concept of CRC .Then understand how to transfer the error free data by applying the XOR	CO2
9.	A category of error detecting (and correcting) code, called the Hamming code, is a code in which dmin = 3. This code can	Apply	Learner to recall the hamming code and then apply &find hamming distance of given code word.	CO2

	detect up to two errors (or correct one single error). In this code, the values of n, k, and r are related as: n = 2r 1 and k = n r. Find the number of bits in the dataword and the codewords if r is 3			
10.	Explain the roles of data link control (DLC) services in ensuring reliable data transmission. What are the main functions of DLC protocols in network communication?	Apply	This would require the learner to recall the concept of DLC. Then understand main functions of its.	CO2
PART-B LONG ANSWER QUESTIONS				
1.	Compare and contrast Go backNandselectiveRepeat	Understand	This would require the learner to recall, analyzethe Data-Link Layer Protocols and flowanderrorcontrol Then compare Go back N and selective Repeat Protocols.	CO2
2.	List and briefly discuss the two different basic transmission technologies.	Understand	—-	CO2
3.	WhatispureALOHAand slotted ALOHA. Consider the delay of both at low load. Whichoneisless? Justify your answer.	Understand	—-	CO2
4.	Summarize the working of carrier sense multiple access protocol	Understand	This would require the learner to recall he concepts of Multiple-access Protocols and Then explain the working of carrier sense multiple access protocol.	CO2
5.	Explaintheback-offtimeof PURE ALOHA protocol	Understand	This would require the learner to recall the conceptsof Random-access protocolsand thenexplain the back- off time of PURE ALOHA protocol	CO2
6.	Describeindetailthetypes of bridges.	Understand	—-	CO2
7.	Explain the functions of MAC.	Understand	This would require the learner to recall the concepts of Media Access Control (MAC) and then explain the functions of MAC	CO2
8.	How performance is improved in CSMA/CD protocol compared to CSMA protocol Explain.	Understand	—-	CO2
9.	How CSMA/CA differs from CSMA/CD. Explain in brief	Understand	—-	CO2

10.	What is the purpose of the timer at the sender site?	Understand	—-	CO2
11.	Explain Error Control & Flow Control.	Understand	This would require the learner to recall the concepts of Network- Layer, Services, protocols: then explain Error Control & Flow Control services	CO2
12.	What we need a multiple access protocol when we use the local loop of the telephone company to access the internet? Explain.	Understand	—-	CO2
13.	Explain about ALOHA and CDMA	Understand	This would require the learner to recall the concepts of Media Access Control	CO2
14.	What is the need for bridges? Explain the working of 802.x to 802.y bridges in detail.	Understand	—-	CO2
15.	What is the need of Flow control? Explain the common approaches for flow control in data link layer	Understand	This would require the learner to recall the concepts of Flow control	CO2
16.	Explain how slotted ALOHA solves the problem of Channel allocation	Understand	This would require the learner to recall the concepts of Media Access Control	CO2
17.	Explain classful and classless addressing schemes with an example?	Understand	This would require the learner to recall the concept of IP addresses	CO2
18.	Explain error detection and correction methods in detail	Understand	This would require the learner to recall the concept of error control	CO2
19.	Explain the services provided by datalinklayer in detail?	Understand	This would require the learner to recall the concept of DLC	CO2
20.	Explain random access protocols in detail?	Understand	This would require the learner to recall the concept of MAC	CO2
PART-C SHORT ANSWER QUESTIONS				
1.	Describe redundancy.	Remember	Learner to define Redundancy Check and then describe the concepts of Cyclic Redundancy Check.	CO2
2.	Define vulnerable period.	Remember	—-	CO2
3.	List out the three categories of multiple access protocols.	Remember	—-	CO2
4.	Define CSMA and CDMA.	Remember	—-	CO2
5.	List out the available error detection methods.	Remember	—-	CO2
6.	What is an exponential back off.	Remember	—-	CO2
7.	What are the responsibilities of data link layer.	Remember	—-	CO2
8.	Explain the different types of	Remember	This would require the learner	CO2

	errors.		to recall the concepts of errors then explaining thedifferent types oferrors.	
9.	Define bridge.	Remember	—-	CO2
10.	Explain Hub.	Remember	This would require the learner to recall the concepts of Hub and Switch then explaining Hub.	CO2
11.	List out the functionalities of router.	Remember	—-	CO2
12.	Define ALOHA.	Remember	—-	CO2
13.	Define checksum.	Remember	—-	CO2
14.	Define VLAN.	Remember	—-	CO2
15.	Explain CRC generator.	Remember	This would require the learner to recall the concepts of detection of a single bit error Then explaining CRC generator.	CO2
16.	How performance is improved in CSMA/CD protocol compared to CSMA protocol.	Remember	—-	CO2
17.	What is vulnerable time?	Remember	—-	CO2
18.	Distinguish between FDMA and TDMA.	Remember	—-	CO2
19.	Explain CRC.	Remember	—-	CO2
20.	Summarize what are the steps followedinchecksum generator.	Remember	This would require the learner to recall the concepts ofdetectionofa singlebiterror andThen explaining checksum generator steps.	CO2
21.	Define parameter 'a'. How does it affect the performance oftheCSMA?	Remember	—-	CO2
MODULE III				
NETWORK LAYER				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1.	Explain the main design issues at the network layer, including addressing, routing, and packet forwarding. How do these issues influence the overall performance and efficiency of a network?	Understand	Learner to recall the concept of network layer and understandtheissuesof that layer.	CO3
2.	Compare and contrast the primary characteristics of link-state and distance-vector routing algorithms. What are the specific advantages and disadvantages of each in various network scenarios?	Understand	—-	CO3
3.	Your organization is	Apply	–	CO6

	expanding its network, and you need to choose a routing algorithm that adapts well to frequent changes in topology. How would you apply the principles of a link-state routing algorithm in this situation? Outline the steps for implementation.			
4.	Find the class of the following IP addresses. a) 237.14.2.1b) 208..35.54.12c) 129.14.6.8 d) 114.34.2.8	Apply	Learner to recall the concept of IPv4 and describe address class. Use the address class and find the class.	CO3
5.	Design the autonomous system with the following specifications : a) There are 8 networks (N1 toN8) b) There are8routers (R1 toR8) c) N1,N2,N3,N4,N5 and N6 are Ethernet LANs d) N7 and N8 are point to point WANs e) R1 connects N1 andN2 f) R2 connects N1 andN7 R3 connects N2 andN8	Understand	Learner to recall the concept of Network and design the autonomous system.	CO3
6.	Consider a host using leaky bucket strategy for traffic shaping. The host sends a burst data at a rate of 15Mbps for first 3 seconds and remains silent for 2 seconds. Then again a burst data at a rate of 6 Mbps is send for next 2 seconds and then the host remains silent for next 2 seconds. Now again the host sends data at rate of 5 Mbps for next 3 seconds. What will be the output data rate of the leaky bucket?	Understand	This would require the learner to recall the concept of traffic shaping. Then explain what is leaky bucket and find the output data rate	CO3
7.	Your organization is running out of available IPv4 addresses due to network growth. Explain why IPv4 addresses are limited and discuss the potential issues that arise from this limitation.	Apply	Learner to recall the concept ofIPV4andIPv6 concept and describe its difference.	CO6
8.	Write an example, demonstrate how to make routing table using distance vector routing. And list down the limitation.	Apply	This would require the learner to recall the concept of routing Algorithms	CO3
9.	Whatarethereasonsfor	Apply	This would require the	CO3

	congestion? Whatarethe problems withcongestion?		learner to recall the concept of congestion control	
10.	Classify the static and dynamic routing algorithms? Explain the basic concept of flooding.	Apply	This would require the learner to recall the concept of routing Algorithms	CO3
11.	What is the format of IPv4 header? Describe the significance of each field.	Apply	This would require the learner to recall the concept of IP Addresses	CO3
12.	A large enterprise network is divided into multiple areas, eachusingOSPFas its routing protocol. How would you configure OSPF toensure efficientrouting between areas and prevent routing loops?	Apply	This would require the learner to recall the concept of OSPF	CO3
13.	What is the need of Flow control? Explain the common approaches for flow control in data link layer.	Apply	This would require the learner to recall the concept of flow control	CO3
14.	Classify the static and dynamic routingalgorithms. Explainthebasicconceptof flooding.	Apply	This would require the learner to recall the concept of routing algorithms	CO3
15.	What is the format of IPv4 header? Describe the significance of each field.	Apply	This would require the learner to recall the concept of IP addresses	CO3
16.	With the growing adoption of IPv6, analyze how transitioningfromIPv4to IPv6impactsthedesignand management of IPnetworks. Whatarethecriticalfactors to consider during the transition?	Apply	This would require the learner to recall the concept of IPv4 and IPv6.	CO3
PART-B LONG ANSWER QUESTIONS				
1.	How the routers get the information about neighbor.	Understand	—-	CO3
2.	How the packet cost is referred in distance vector and link state routing.	Understand	—-	CO3
3.	Describe the Routing Information protocol and Distance vector routing protocol.	Understand	This would require the learner to recall the concept of Unicast Routing Protocols and Routing Algorithms and then describe the Routing Information protocol and Distance vector routing protocol.	CO3
4.	Explain Leaky bucket algorithm	Understand	This would require the learner to recall theconcept of Traffic Shaping or Policing andthenexplain theLeaky bucketalgorithm	CO3

5.	Explain the Traffic Shaping	Understand	This would require the learner to recall the concept of Flow Control To Improve QoS and then explain the Traffic Shaping or Policing	CO3
6.	Illustrateindetailabout non-adaptivealgorithms	Understand	This would require the learner to recall the concept of non-adaptive algorithms and thenexplaintheabout non-adaptive algorithms	CO3
7.	ExplaintheFlooding algorithms	Understand	This would require the learner to recall the concept of Flooding and then explain the Flooding algorithms	CO3
8.	List the fields of an IPv4 datagram header that participate in fragmentation and reassembly.	Understand	—-	CO3
9.	Demonstrate the link state routing algorithm with an example	Understand	This would require the learnertorecalltheconcept of UnicastRoutingandthen explain the link state routing algorithmwithan example	CO3
10.	State the major difference between Distance Vector Routing and Link state routing. Discuss	Understand	—-	CO3
11.	Explain the various congestion control mechanism in detail	Understand	This would require the learner to recall the concept of congestion control and then explain The various congestion control mechanism in detail	CO3
12.	Explain Internet Protocol withtheneatblockdiagram of IP headerformat.	Understand	This would require the learner to recall the concept of Network-Layer Protocols and then draw and explain the neat block diagram of IP header format.	CO3
13.	List and explain the features of the IPv6 Protocol.	Understand	—-	CO3
14.	Illustrate the IP packet format with neat diagram.	Understand	This would require the learnertorecalltheconcept ofIPv6 Protocolandexplain the IPpacket format with neat diagram.	CO3
15.	Explain the IPv6packet format.	Understand	—-	CO3
16.	Discuss the datagram delivery and forwarding in internet protocol.	Understand	This would require the learner to recall the concept of datagram delivery and forwarding in internet protocol and then explain datagram delivery and forwarding.	CO3

17.	What is the purpose of routing algorithms in the network layer?	Understand	—-	CO3
18.	Define IPv6 protocol.	Understand	—-	CO3
19.	Explain about Internet Control Message Protocol.	Understand	This would require the learnertorecalltheconcept of Internet Protocol and explain the Internet Control Message Protocol.	CO3
20.	Explain about IP addressing methods.	Understand	This would require the learner to recall the concept of IP addressing methods and explain the IP addressing methods.	CO3
21.	Classify two groups of multicast routing protocols.	Understand	This would require the learner to recall the concept of multicast routing protocols and Classify two groups of multicast routing protocols.	CO3
PART-C SHORT ANSWER QUESTIONS				
1.	Explain quality of service.	Remember	This would require the learner to recall the concept of internetworking issue and then explainthequalityof service	CO3
2.	List the classifications of the adaptive algorithms.	Remember	—-	CO3
3.	List the classifications of the non- adaptivealgorithms.	Remember	—-	CO3
4.	Define the keys for understanding the distance vector routing.	Remember	—-	CO3
5.	Define Flooding.	Understand	—-	CO3
6.	What is meant byrouting algorithm?	Remember	—-	CO3
7.	Illustrate optimality principle.	Remember	—-	CO3
8.	Define Adaptive routing algorithms.	Remember	—-	CO3
9.	Define Non-Adaptive routing algorithms.	Remember	—-	CO3
10.	What is congestion control?	Understand	—-	CO3
11.	Define Traffic shaping.	Remember	—-	CO3
12.	Explain Leaky bucket algorithm.	Remember	This would require the learner to recall the concept of Leaky bucket algorithm and then explain the Leaky bucket algorithm	CO3
13.	Define Load shedding.	Understand	—-	CO3
14.	What are thedesign issues of network layer.	Understand	—-	CO3
15.	List network support layers and the user support layers.	Understand	—-	CO3
16.	Explainstoreandforward.	Understand	This would require the learner to recall the concept of	CO3

			store and forward and then explain store and forward	
17.	Define shortest path.	Understand	—-	CO3
18.	List out the keys for understanding the link state routing.	Remember	—-	CO3
19.	Listtherequirementsofthe routing algorithms.	Remember	—-	CO3
20.	List the three variants of the internetworking.	Remember	—-	CO3
21.	Define virtual circuit.	Understand	—-	CO3
22.	List out responsibilities of network layer.	Understand	—-	CO3
23.	Define datagram.	Remember	—-	CO3
24.	How broadcast and multicast address is represented in IP addressing scheme?	Remember	—-	CO3
25.	Distinguish between Datagram and datagram networks.	Understand	This would require the learner to recall the concept of datagram then explain the differences between Datagram and datagram networks.	CO3
26.	Define IPv4.	Remember	—-	CO3
27.	List out functions of IP.	Remember	—-	CO3
MODULE IV				
TRANSPORT LAYER				
PART A- PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1.	Assume An end system sends 50 packets for second using UDP over a full duplex mode 100 Mbps Ethernet LAN Connection. Each packet consists of 1500 Bytes of the Ethernet frame payload data. What is the throughput when measured at UDP protocol?	Apply	This would require the learner to recall the concept of IP. understand the concept of UDP protocol and then find the Packets of UDP over a full duplex mode and then applying the throughput.	CO4
2.	Imagine you are developing an online multiplayer game. Explain how the choice between TCP and UDP could affect the gameplay experience. Why might one protocol be preferred over the other in this scenario?	Understand	This would require the learner to recall the concept of IP. understand the concept of UDP protocol and TCP protocol.	CO6
3.	Design a TCP connection is using a window size of 12000 bytes and the previous acknowledgement remembrance number was 22001.It receives a segment with acknowledgment	Understand	This would require the learner to recall the concept of TCP and Then design a diagram to show the situation of the window before and after.	CO6

	number 24001 and window size advertisement of 12000. Design a diagram to show the situation of the window before and after.			
4.	A video streaming serviceis struggling with congestion on its network during peak usage times. Explain the concept of congestion control in the transport layer and describe how it can help alleviate network congestion in this context.	Understand	This would require the learner to recall the concept of IP. understand the concept of UDP protocol and TCP protocol.	CO4
5.	You are tasked with developing a file transfer application that ensures reliable delivery of files over an unreliable network. How would you use TCP to implement this reliability? Discuss the features of TCP that would be Beneficial for this application.	Apply	This would require the learner to recall the concept of TCP and Then design a diagram to show the situation of the window before and after.	CO6
6.	A company's internal network is experiencing slow file transfers despite having high bandwidth. How would you tune TCP settings to improve performance? Explain the steps you would take and the TCP parameters you would adjust.	Apply	This would require the learner to recall & relate the concept of UDP and Then apply to Calculate the efficiency of this transmission at the UDP level.	CO4
7.	Discuss in detail about the connection establishment and release in TCP.	Apply	This would require the learner to recall the concept of TCP and Then design a diagram to show the situation of the window before and after.	CO4
8.	Describe in detail about TCP segment header and connection Establishment.	Apply	This would require the learner to recall the concept of TCP and Then design a diagram to show the situation of the window before and after.	CO4
9.	a) Explain the Services of Transport layer. b) Explain leaky bucket and token bucket algorithms	Apply	This would require the learner to recall the concept of Transport layer services and congestion control	CO4

10.	Draw and explain each field in the TCP Segment header.	Apply	This would require the learner to recall the concept of TCP and Then design a diagram to show the situation of the window before and after.	CO4
PART-B LONG ANSWER QUESTIONS				
1.	Explain the real transport protocol of UDP and how will you calculate checksum in UDP.	Apply	This would require the learner to recall the concept of transport protocol of UDP and Then explain real transport protocol of UDP and how will you calculate checksum in UDP	CO4
2.	Show neatly the TCP segment format and describe each of it.	Apply	—-	CO4
3.	List out the network performance characteristics.	Understand	—-	CO4
4.	Illustrate the adaptive retransmission policy in detail.	Apply	This would require the learner to recall the concept of adaptive retransmission policy and Then explain adaptive retransmission policy in detail.	CO4
5.	Show the TCP connection establishment and termination using timeline diagram.	Understand	—-	CO4
6.	Explain the three way handshake protocol to establish the transport level connection.	Apply	—-	CO4
7.	Design TCP state transition diagram and describe each of it.	Apply	This would require the learner to recall the concept of TCP state transition diagram and Then explain Design TCP state transition diagram and Describe each of it.	CO4
8.	Explain a detailed note on connection establishment.	Apply	This would require the learner to recall the concept of connection establishment and then explain detailed note on connection establishment.	CO4
9.	Discuss about the TCP sliding window algorithm for flow control.	Understand	This would require the learner to recall the concept of TCP sliding window and Then explain TCP sliding window algorithm for flow control.	CO4

10.	Summarize all congestion control algorithms and describe how it works.	Apply	This would require the learner to recall the concept of congestion control algorithm and Then explain congestion control algorithms and describe how it works	CO4
11.	Illustrate leaky bucket and token bucket algorithm.	Apply	This would require the learner to recall the concept of leaky bucket and token bucket and Then explain leaky bucket and token bucket algorithm.	CO4
12.	Compare & Contrast UDP & TCP with suitable example.	Apply	This would require the learner to recall the concept of UDP & TCP and Then explain the differences of UDP & TCP	CO4
13.	Explain congestion avoidance techniques in detail.	Understand	This would require the learner to recall the concept of congestion avoidance techniques and Then explain the congestion avoidance techniques indetail.	CO4
14.	List major types of networks and give brief note on each of it.	Understand	—-	CO4
15.	Illustrate data units at different layers of the TCP / IP protocol suite.	Apply	This would require the learner to recall the concept of data units and Then explain data units at different layers of the TCP / IP protocol suite.	CO4
16.	Discuss in detail about the connection establishment and release in TCP.	Apply	This would require the learner to recall the concept of data units and Then explain data units at different layers of the TCP / IP protocol suite.	CO4
17.	Draw and explain each field in the TCP Segment header.	Apply	This would require the learner to recall the concept of data units and Then explain data units at different layers of the TCP / IP protocol suite.	CO4
18.	Explain leaky bucket and token bucket algorithms.	Apply	This would require the learner to recall the concept of Congestion control Algorithms	CO4
19.	Explain in detail about transport layer protocols	Understand	This would require the learner to recall the concept of Transport layer	CO4
20.	Explain the services provided by the transport layer	Understand	This would require the learner to recall the concept of Transport layer	CO4

PART-C SHORT ANSWER QUESTIONS				
1.	List out the functions of transport layer.	Remember	—-	CO4
2.	Define Multi-protocol router.	Remember	—-	CO4
3.	List out duties of the transport layer.	Remember	—-	CO4
4.	Define role of TCP in networks.	Remember	—-	CO4
5.	Distinguish between network layer delivery and the transport layer delivery.	Remember	This would require the learner to recall the concept of network layer delivery and	CO4
			transport layer delivery Then explain & contrast the Differences of network layer delivery and the transport layer delivery.	
6.	What are the different fields in pseudo header?	Remember	—-	CO4
7.	Define quality of service.	Understand	—-	CO4
8.	What is the main idea of UDP.	Remember	—-	CO4
9.	List the timers used by TCP.	Remember	—-	CO4
10.	How an application process running in one host is addressed by another process through TCP.	Remember	—-	CO4
11.	Show datagram format of UDP.	Remember	This would require the learner to recall the concept of UDP and explain the datagram format of UDP.	CO4
12.	What is traffic shaping?	Remember	—-	CO4
13.	What are the two protocols available at transport layer?	Remember	—-	CO4
14.	List out various congestion avoidance techniques.	Understand	—-	CO4
15.	Distinguish between contention and congestion.	Understand	—-	CO4
16.	Define tunneling.	Understand	—-	CO4
17.	State the four major aspects of reliable delivery at the transport layer.	Remember	—-	CO4
18.	How check sum is calculated in TCP.	Remember	—-	CO4
19.	What is CODE BITS in TCP header?	Remember	—-	CO4
20.	Define the use of SYN and FIN bits in TCP.	Remember	—-	CO4
21.	What is the difference between TCP?	Understand	—-	CO4
22.	Show UDP header format.	Remember	—-	CO4
23.	Explain a short note on transport layer services.	Remember	This would require the learner to recall the concept of transport layer and Then explain the transport layer services.	CO4

24.	What is congestion? How to control congestion.	Understand	—-	CO4
25.	Define multiplexing.	Remember	—-	CO4
26.	How connection establishment is acquiring.	Remember	—-	CO4
27.	How to release a connection from the network.	Remember	—-	CO4
28.	Define Internet Transport Protocols.	Understand	—-	CO4
MODULE V				
THE LINK LAYER LAYER and LANs				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS)				
1.	Explain the services provided by the Link Layer.	Understand	This requires learners to recall basic link layer concepts and apply their understanding to explain the services provided.	CO5
2.	Illustrate the working of the DOCSIS protocol for cable internet access.	Understand	Requires learners to recall and apply knowledge of DOCSIS, explaining its role in cable internet access.	CO5
3.	Differentiate between Frame Switching and Packet Switching.	Understand	Helps learners analyze key differences and apply switching concepts to networking scenarios.	CO5
4.	Determine the advantages of Multiprotocol Label Switching (MPLS) over traditional IP routing.	Understand	Encourages critical thinking by comparing MPLS and IP routing efficiency.	CO5
5.	Interpret the concept of Link Virtualization and its impact on modern networking.	Apply	Enhances understanding of virtualization and its role in modern networks.	CO5
6.	Describe how a Data Centre network is structured and its significance in cloud computing.	Understand	Helps learners understand cloud networking by analyzing data centre structures.	CO5
7.	Compare the functionality of MAC addressing and IP addressing in Link Layer communication.	Understand	Develops analytical skills by comparing data link and network layer addressing.	CO5
8.	Explain the working mechanism of ARP and RARP protocols.	Understand	Reinforces knowledge of address resolution in network communications.	CO5
9.	What are the main error detection techniques used in the Link Layer?	Understand	Helps learners understand key techniques in maintaining data integrity.	CO5
10.	Illustrate the role of VLANs in modern networking and their implementation in Data Centre networks.	Understand	Reinforces network segmentation concepts for efficient traffic management.	CO5
PART-B LONG ANSWER QUESTIONS				
1.	Describe the different types of services provided by the Link Layer.	Understand	Strengthens understanding of core link-layer functionalities.	CO5

2.	Explain the architecture and components of DOCSIS.	Understand	Develops an in-depth understanding of DOCSIS implementation.	CO5
3.	Discuss the benefits and challenges of implementing Link Virtualization.	Understand	Enables learners to analyze virtualization in networking.	CO5
4.	Illustrate the working principle of MPLS and its role in modern networking.	Understand	Strengthens conceptual clarity on MPLS and routing techniques.	CO5
5.	Compare the networking needs of a traditional enterprise network versus a data centre network.	Understand	Provides insights into enterprise vs. cloud networking.	CO5
6.	How does MPLS enhance network performance compared to conventional routing techniques?	Understand	Encourages a comparative analysis of routing methods.	CO5
7.	Describe the role of Software-Defined Networking (SDN) in data centres.	Understand	Helps learners analyze SDN concepts for modern networks.	CO5
8.	Explain the Spanning Tree Protocol (STP) and its importance in preventing loops.	Understand	Encourages in-depth evaluation of DNS and HTTP protocols.	CO5
9.	Illustrate the significance of Network Address Translation (NAT) in network communication.	Understand	Develops understanding of TCP/IP data transfer mechanisms.	CO5
10.	What is a collision domain and how does switching help in reducing it?	Understand	Explains the importance of access control in authentication.	CO5
11.	Compare and contrast Layer 2 and Layer 3 switches.	Understand	Helps learners differentiate between transport layer and application layer security.	CO5
12.	Explain the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) mechanism.	Understand	Allows learners to analyze the role of addressing schemes in network communication.	CO5
13.	Describe the role of Ethernet frames in data link communication.	Understand	Enables students to apply security mechanisms in a layered network model.	CO5
14.	Explain the process of Virtual LAN (VLAN) tagging.	Understand	Strengthens understanding of email protocols and message transmission mechanisms.	CO5
15.	Discuss how Load Balancing is achieved in Data Centre Networking.	Understand	Develops comprehension of how Layer 2 and Layer 3 networking devices function.	CO5
16.	Illustrate the functions of Layer 2 Tunneling Protocol (L2TP) in VPNs.	Understand	Illustrates the importance of QoS in networking.	CO5

17.	Describe the differences between MPLS and traditional IP routing.	Understand	Encourages evaluation of traditional networking against modern network virtualization.	CO5
18.	Explain the significance of Ethernet in wired networking.	Understand	Helps learners apply the principles of link-layer transmission in different protocols.	CO5
19.	Describe the major features of FDDI networks.	Understand	Improves knowledge of switching protocols and data transfer methods.	CO5
20.	Discuss the major	Understand	Explains the practical	CO5
	enhancements in 5G networking that impact Data Centre Networks.		applications of data link-layer functionalities.	
PART-C SHORT ANSWER QUESTIONS				
1.	What is the purpose of the Link Layer in the OSI model?	Remember	Reinforces understanding of cloud data centre connectivity.	CO5
2.	List the main functions of DOCSIS in cable internet access.	Remember	Explains how VLANs support network security and efficiency.	CO5
3.	Define Link Virtualization and its advantages.	Remember	Encourages evaluation of error detection techniques in data transmission.	CO5
4.	What are the key differences between traditional routing and MPLS?	Remember	Enhances practical knowledge of transmission protocols in network communication.	CO5
5.	What is a Data Centre Network, and why is it important?	Remember	Demonstrates the impact of networking techniques on large-scale data networks.	CO5
6.	Explain the concept of Layer 2 and Layer 3 switching.	Remember	Helps learners differentiate between control and data plane functionalities in networks.	CO5
7.	List two key protocols used in Data Centre Networking.	Remember	Allows students to assess the practical challenges in wired and wireless networks.	CO5
8.	Define MPLS labels and their role in packet forwarding.	Remember	Improves comprehension of device addressing and topology formation.	CO5
9.	How does a data centre network differ from a traditional network?	Remember	Encourages understanding of hierarchical and flat addressing models in networking.	CO5
10.	What is the significance of Quality of Service (QoS) in MPLS?	Remember	This requires learners to recall basic link layer concepts and apply their understanding to explain the services provided.	CO5
11.	Explain the basics of Ethernet frame structure.	Remember	Requires learners to recall and apply knowledge of DOCSIS, explaining its role in cable internet access.	CO5

12.	Define the term 'packet fragmentation' and how it affects networking.	Remember	Helps learners analyze key differences and apply switching concepts to networking scenarios.	CO5
13.	List the benefits of using VLANs in a large network.	Remember	Encourages critical thinking by comparing MPLS and IP routing efficiency.	CO5
14.	What is the function of an access point in a wireless network?	Remember	Enhances understanding of virtualization and its role in modern networks.	CO5
15.	Describe the role of Data Centre Interconnect (DCI).	Remember	Helps learners understand cloud networking by analyzing data centre structures.	CO5
16.	How does a load balancer work in a network?	Remember	Develops analytical skills by comparing data link and network layer addressing.	CO5
17.	Explain the concept of Network Function Virtualization (NFV).	Remember	Reinforces knowledge of address resolution in network communications.	CO5
18.	What is the function of MAC learning in a switch?	Remember	Helps learners understand key techniques in maintaining data integrity.	CO5
19.	List the advantages of Software-Defined Networking (SDN).	Remember	Reinforces network segmentation concepts for efficient traffic management.	CO5
20.	Describe the importance of IPsec in secure communication.	Remember	Strengthens understanding of core link-layer functionalities.	CO5

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